

Week3-practical

Task3:

Develop a C program using `fork()` that creates a parent and child process. Display the Process ID (PID), Parent Process ID (PPID), and process states at different stages of execution.

```
UW PIDO 5.09 File: task3.c Modif

#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>

int main() {
    pid_t pid;

    printf("Before fork()\n");
    printf("Current PID: %d\n", getpid());

    pid = fork();

    if (pid < 0) {
        perror("fork failed");
        exit(1);
    }

    if (pid == 0) {
        // Child process
        printf("Child Process\n");
        printf("PID = %d\n", getpid());
        printf("PPID = %d\n", getppid());
        printf("State: Running\n");

        sleep(10);

        printf("Child exiting...\n");
        exit(0);
    }

    else {
        // Parent process
        printf("Parent Process\n");
        printf("PID = %d\n", getpid());
        printf("PPID = %d\n", getppid());
        printf("Child PID = %d\n", pid);
        printf("State: Running\n");

        printf("Parent waiting for child...\n");
        wait(NULL);

        printf("Child terminated.\n");
        printf("Parent exiting...\n");
    }

    return 0;
}

Get Help Exit WriteOut Justify Read File Where is Prev Pg Next Pg Cut Text UnCut Text Cur Pos To Spell

Last login: Mon Aug 17 20:01:12 on ttys001
(eval):8: command not found: x
vishalkarnati@VISHALS-MacBook-Air ~ % cd ~/Desktop/os
vishalkarnati@VISHALS-MacBook-Air os % mkdir week3pract
vishalkarnati@VISHALS-MacBook-Air os % cd week3pract
vishalkarnati@VISHALS-MacBook-Air week3pract % nano task3.c
vishalkarnati@VISHALS-MacBook-Air week3pract % gcc task3.c -o task3
vishalkarnati@VISHALS-MacBook-Air week3pract % ./task3
Before fork()
Current PID: 7948

Parent Process
PID = 7948
PPID = 7921
Child PID = 7949
State: Running
Parent waiting for child...
Child Process
PID = 7949
PPID = 7948
State: Running
Child exiting...
Child terminated.
Parent exiting...
vishalkarnati@VISHALS-MacBook-Air week3pract %

Last login: Mon Aug 17 20:11:56 on ttys000
(eval):8: command not found: x
vishalkarnati@VISHALS-MacBook-Air ~ % ps -el | grep task3
501 7966 7951 4006 0 31 0 410059936 224 ~ R+ 0 ttys001 0:00.00 grep task3
vishalkarnati@VISHALS-MacBook-Air ~ %
```

Task4:

Design an experiment to observe process state transitions (Ready, Running, Waiting, Terminated) using Linux monitoring tools such as `ps`, `top`, and `/proc`. Document the observations.

```
UW PICO 5.09 File: task4.c Modified
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

int main() {
    pid_t pid = fork();

    if (pid < 0) {
        perror("fork failed");
        exit(1);
    }

    if (pid == 0) {
        printf("Child Process\n");
        printf("Child PID: %d\n", getpid());
        printf("Child PPID: %d\n", getppid());

        printf("Child is running...\n");

        // Keep child alive so we can observe it
        sleep(15);

        printf("Child terminating...\n");
        exit(0);
    }
    else {
        printf("Parent Process\n");
        printf("Parent PID: %d\n", getpid());
        printf("Child PID: %d\n", pid);

        printf("Parent waiting for child...\n");
        wait(NULL);

        printf("Child terminated.\n");
        printf("Parent terminating...\n");
    }

    return 0;
}

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X Exit B Justify W Where is A Next Pg U UnCut Text A To Spell

vishalkarnati@VISHALS-MacBook-Air week3pract % nano task4.c
vishalkarnati@VISHALS-MacBook-Air week3pract % gcc task4.c -o task4
vishalkarnati@VISHALS-MacBook-Air week3pract % ./task4
Parent Process
Parent PID: 7994
Child PID: 7995
Parent waiting for child...
Child Process
Child PID: 7995
Child PPID: 7994
Child is running...
Child terminating...
Child terminated...
Parent terminating...
vishalkarnati@VISHALS-MacBook-Air week3pract %

Last login: Mon Aug 17 20:17:35 on ttys002
(eval):8: command not found: >
vishalkarnati@VISHALS-MacBook-Air ~ % ps aux | grep task4
vishalkarnati  8052  0.0  0.0 435299712 1328 s001 S+   8:18PM   0:00.00 grep task4
vishalkarnati@VISHALS-MacBook-Air ~ % ps -p PID -o pid,ppid,state,command
ps: Invalid process id: PID
vishalkarnati@VISHALS-MacBook-Air ~ % top
vishalkarnati@VISHALS-MacBook-Air ~ %
```

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